In-Class 3.4 — Completing the Square — Answer Key

Math 103: College Algebra

Completing the Square

Fill in the blanks to complete the square. Use the squares provided.

1. $x^2 + 8x + 16 = (x+4)^2$

x^2	$4x$
$4x$	16

2. $x^2 - 4x + 4 = (x + (-2))^2$

x^2	$-2x$
$-2x$	4

3. $x^2 - 7x + \frac{49}{4} = \left(x + \frac{7}{2}\right)^2$

x^2	$\frac{7}{2}x$
$\frac{7}{2}x$	$\frac{49}{4}$

4. $x^2 + \frac{7}{5}x + \frac{49}{100} = \left(x + \frac{7}{10}\right)^2$

x^2	$\frac{7}{10}x$
$\frac{7}{10}x$	$\frac{49}{100}$

Solve by Completing the Square

Solve each quadratic equation by completing the square.

5.
$$\begin{aligned} x^2 - 4x + 4 &= 8 + 4 \\ (x-2)^2 &= 12 \\ x - 2 &= \pm\sqrt{12} \\ x - 2 &= \pm 2\sqrt{3} \\ x &= 2 \pm 2\sqrt{3} \end{aligned}$$

6.
$$\begin{aligned} x^2 + 3x + \frac{9}{4} &= 5 + \tfrac{9}{4} \\ \left(x + \tfrac{3}{2}\right)^2 &= \tfrac{29}{4} \\ x + \tfrac{3}{2} &= \pm\sqrt{\tfrac{29}{4}} \\ x + \tfrac{3}{2} &= \pm\tfrac{\sqrt{29}}{2} \\ x &= -\tfrac{3}{2} \pm \tfrac{29}{2} \\ x &= \tfrac{-3 \pm 29}{2} \end{aligned}$$

7.
$$\begin{aligned} x^2 - 4x &= -3 \\ x^2 - 4x + 4 &= -3 + 4 \\ (x-2)^2 &= 1 \\ x - 2 &= \pm\sqrt{1} \\ x &= 2 \pm 1 \\ x &= 1,\ x = 3 \end{aligned}$$

8.
$$\begin{aligned} x^2 - 7x &= 16 \\ x^2 - 7x + \tfrac{49}{4} &= 16 + \tfrac{49}{4} \\ \left(x - \tfrac{3}{2}\right)^2 &= \tfrac{113}{4} \\ x - \tfrac{3}{2} &= \pm\sqrt{\tfrac{113}{4}} \\ x - \tfrac{3}{2} &= \pm\tfrac{\sqrt{113}}{2} \\ x &= \tfrac{3}{2} \pm \tfrac{\sqrt{113}}{2} \\ x &= \tfrac{3 \pm \sqrt{113}}{2} \end{aligned}$$

9.
$$\begin{aligned} 2x^2 + 8x &= 10 \\ x^2 + 4x &= 5 \\ x^2 + 4x + 4 &= 5 + 4 \\ (x+2)^2 &= 9 \\ x + 2 &= \pm\sqrt{9} \\ x + 2 &= \pm 3 \\ x &= -2 \pm 3 \\ x &= -5,\ x = 1 \end{aligned}$$

10.
$$\begin{aligned} -\tfrac{2}{3}x^2 + 6x &= -4 \\ x^2 - 9x &= 6 \\ x^2 - 9x + \tfrac{81}{4} &= 6 + \tfrac{81}{4} \\ \left(x - \tfrac{9}{2}\right)^2 &= \tfrac{105}{4} \\ x - \tfrac{9}{2} &= \pm\sqrt{\tfrac{105}{4}} \\ x - \tfrac{9}{2} &= \pm\tfrac{\sqrt{105}}{2} \\ x &= \tfrac{9}{2} \pm \tfrac{\sqrt{105}}{2} \\ x &= \tfrac{9 \pm \sqrt{105}}{2} \end{aligned}$$

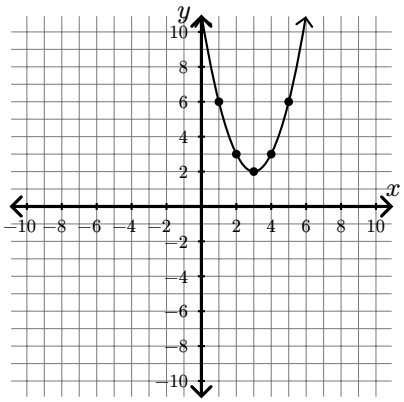
Graphing Parabolas in Vertex Form

For each quadratic function below:

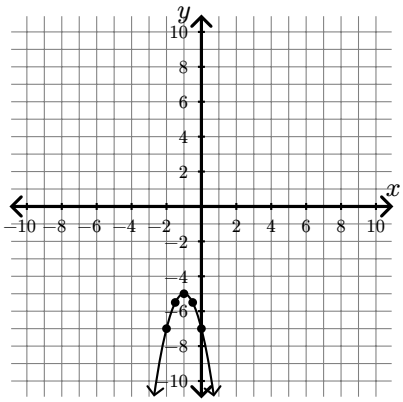
a. Identify and plot the vertex.

b. Plot two symmetric points with integer coordinates on each side of the vertex.

11.



12.



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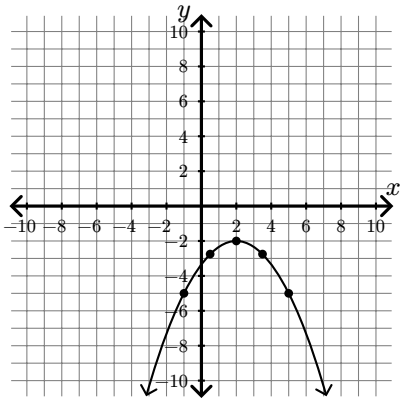

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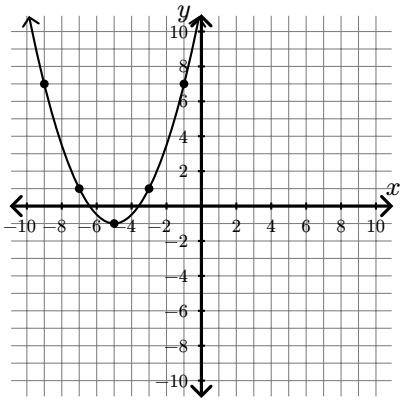


Graphing by Rewriting in Vertex Form

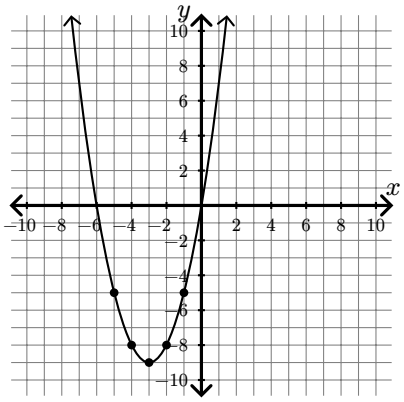
For each quadratic function below:

a. Rewrite the function in vertex form.

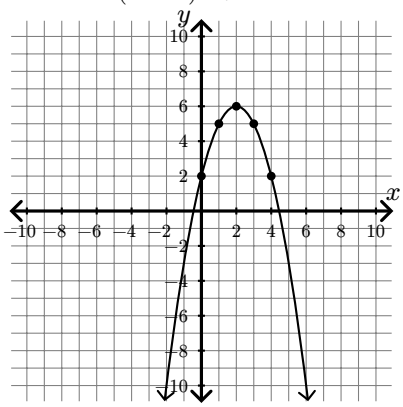
b. Identify and plot the vertex.

c. Plot two symmetric points with integer coordinates on each side of the vertex.

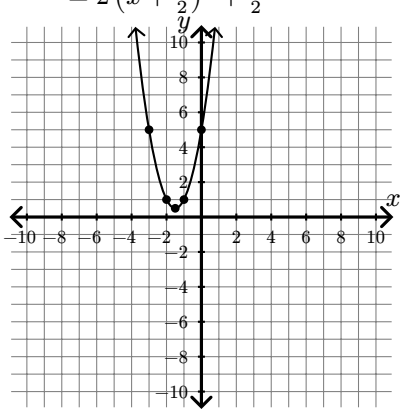
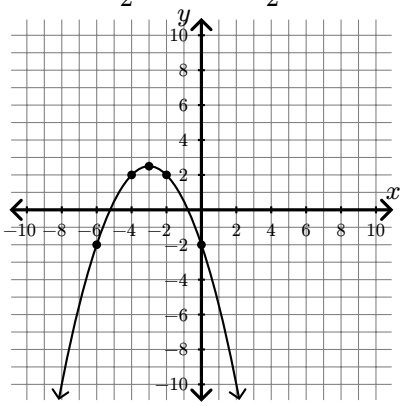
15. $f(x) = x^2 - 6x$
$= x^2 + 6x + 9 - 9$
$= (x+3)^2 - 9$



16. $f(x) = -x^2 + 4x + 2$
$= -\left(x^2 - 4x\right) + 2$
$= -\left(x^2 - 4x + 4 - 4\right) + 2$
$= -\left((x-2)^2 - 4\right) + 2$
$= -(x-2)^2 + 4 + 2$
$= -(x-2)^2 + 6$



17. $f(x) = 2x^2 + 6x + 5$
$= 2\left(x^2 + 3x\right) + 5$
$= 2\left(x^2 + 3x + \frac{9}{4} - \frac{9}{4}\right) + 5$
$= 2\left(\left(x + \frac{3}{2}\right)^2 - \frac{9}{4}\right) + 5$
$= 2\left(x + \frac{3}{2}\right)^2 - \frac{9}{2} + 5$
$= 2\left(x + \frac{3}{2}\right)^2 + \frac{1}{2}$



18. $f(x) = -\dfrac{1}{2}x^2 - 3x - 2$
$= -\dfrac{1}{2}\left(x^2 - 6x\right) - 2$
$= -\dfrac{1}{2}\left(x^2 - 6x + 9 - 9\right) - 2$
$= -\dfrac{1}{2}\left((x-3)^2 - 9\right) - 2$
$= -\dfrac{1}{2}(x-3)^2 + \dfrac{9}{2} - 2$
$= -\dfrac{1}{2}(x-3)^2 + \dfrac{5}{2}$